MATH 170: THE LANGUAGE OF ARTIFICIAL INTELLIGENCE

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Math is the basic language underpinning the algorithms and models used in artificial intelligence. The course is designed to give a gentle introduction of that language, demystifying mathematical vocabulary and concepts in order to understand the usage and application of AI in society and daily life. The course is designed for students from all backgrounds and knowledge in math, and will emphasize mathematical literacy and communication skills.

TOPICS

- Intro to the math of AI: supervised, unsupervised & reinforcement learning
- Points, lines & algebra review
- Datasets and plots
- Teachable Machines
- Binary vectors and character recognition
- Vectors, distance, and scaling
- Nearest-neighbors algorithm
- Regression
- k-Means clustering
- Logic
- Decision trees
- Graphs and networks
- Graphs and Matrices
- Markov processes
- Matrices and Vectors
- Calculus and Optimization
- Neural Nets
- Reinforcement Learning

BASIC COURSE INFO

- Open to anyone, any major!
- No prerequisites
- No exams
- In-person
- Satisfies Gen-Ed

GEN AI ASSISTANCE FOR LESSON PREP

- Create fictional datasets
- Quickly generate multiple-choice/quick assessments
- Generate multiple parallel fictional scenarios
- Create engaging (not scientific) illustrations on the fly to make activities more accessible

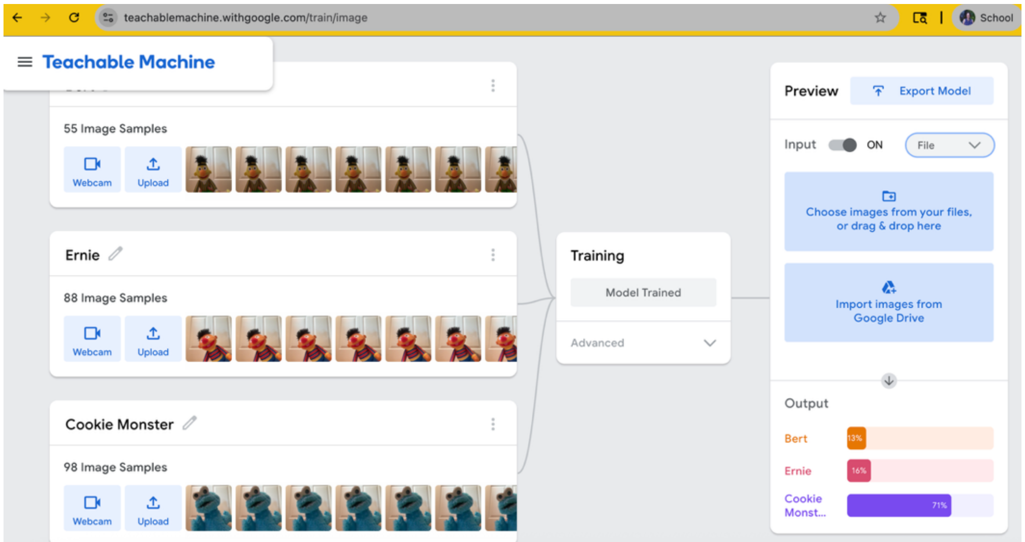
+ Gen AI is useful tool for developing narrative-based activities

Sample Activity 1: **Grocery Shoppers and Decision Trees**

Goal: Teams construct a decision tree to classify grocery stoppers into one of 6 age-related demographics based on the contents of their grocery store purchases.

- Gen AI can quickly generate fictional shopping carts
- Cartoon images help quickly convey the categories

Sample Activity 2: **Teachable Machines**



https://teachablemachine.withgoogle.com/

- Train a machine learning model on images uploaded with a webcam
- No machine learning expertise or coding required
- Conveys the idea of test & training data, biased data, accuracy measures

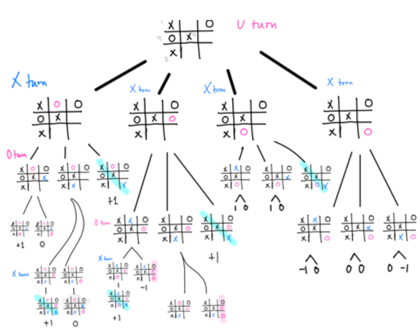
Sample Activity 2: **Practice logical statements and negation with Gemini**

"I am a student and my goal is to practice negation with exercises that help me understand logical and conditional statements. I would like for you to make me an 8 question, multiple choice quiz. Each question should present a slightly different type of logical statement and should ask me to correctly negate it. Include examples with conjunctions, disjunctions, universal and existential quantifiers and complex implications. Every sentence should be in English only, with no symbols, and do not emphasize keywords."

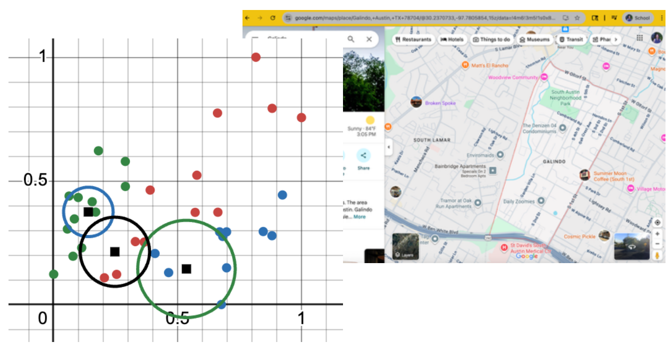
- Provide the prompt
- Gemini generates a quiz
- After practicing, students take graded quiz

Sample Activity 3: **Minimax algorithm with Tic-Tac-Toe**

Work together at the whiteboard to understand an algorithm commonly used by AIs in two-player games.



Sample Activity 4: **Clustering Algorithms and Regression**



- Clustering: ML and Stats techniques for data analysis
- Scale numeric data
- Develop plots in Desmos
- Predict house location and pricing
- Observe concrete effects of biased data

Soft skills:

- using Excel and Google spreadsheets
- downloading and opening files in the correct program
- using Desmos at an intermediate-level to make plots